VI SEMESTER (PROFESSIONAL ELECTIVE-II)	L	T	P	C
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20CS6E01 - SOFTWARE TESTING METHODOLOGIES				

COURSE OBJECTIVES:

1. To study fundamental concepts in software testing and discuss various software testing issues and solutions in software unit, integration, regression and system testing
2. To learn how to plan a test project, design test cases and data, conduct testing, manage software problems and defects, generate a test report
3. To expose the advanced software testing concepts such as object-oriented software testing methods, web-based and component-based software testing
4. To understand software test automation problems and solutions
5. To learn how to write software test documents and communicate with engineers in various forms

COURSE OUTCOMES:

By the end of the course, the student should have the ability to:

CO1: Identify and understand various software testing problems, apply software testing knowledge and engineering methods and solve these problems by designing and selecting software test models, criteria, strategies, and methods

CO2: Design and conduct a software test process for a software project

CO3: Analyze the needs of software test automation

CO4: Use various communication methods and skills to communicate with their teammates to conduct their practice-oriented software testing projects

CO5: Basic understanding and knowledge of contemporary issues in software testing, such as component-based, web based and object oriented software testing problems

CO6: Write test cases for given software to test it before delivery to the customer and write test scripts for both desktop and web based applications

UNIT-I:

Software Testing: Introduction, Evolution, Myths & Facts, Goals, Psychology, definition, Model for testing, Effective Vs Exhaustive Software Testing. Software Testing Terminology and Methodology: Software Testing Terminology, Software Testing Life Cycle, Software Testing Methodology. Verification and **Validation:** Verification & Validation Activities, Verification, Verification of Requirements, High level and low level designs, verifying code, Validation.

UNIT-II:

Dynamic Testing-Black Box testing techniques: Boundary Value Analysis, Equivalence class Testing, State Table based testing, Decision table based testing, Cause-Effect Graphing based testing, Error guessing.

White-Box Testing: need, Logic Coverage criteria, Basis Path testing, Graph matrices, Loop testing, data flow testing, mutation testing.

UNIT-III:

Static Testing: Inspections, Structured Walkthroughs, Technical Reviews.

Validation activities: Unit testing, Integration Testing, Function testing, system testing, acceptance testing.

Regression testing: Progressives Vs regressive testing, Regression test ability, Objectives of regression testing, Regression testing types, Regression testing techniques.

UNIT-IV:

Efficient Test Suite Management: growing nature of test suite, Minimizing the test suite and its benefits, test suite prioritization, Types of test case prioritization, prioritization techniques, measuring the effectiveness of a prioritized test suite ,Software Quality Management: Software Quality metrics, SQA models. Debugging: process, techniques, correcting bugs.

UNIT V

Automation and Testing Tools: need for automation, categorization of testing tools, selection of testing tools, Cost incurred, Guidelines for automated testing, overview of some commercial testing tools such as Win Runner, Load Runner, Jmeter and JUnit . Test Automation using Selenium tool.

Testing Object Oriented Software: basics, Object oriented testingTesting Web based Systems: Challenges in testing for web based software, quality aspects, web engineering, testing of web based systems.

Text Books:

1) Software Testing, Principles and Practices, NareshChauhan, Oxford
2) Software Testing, Yogesh Singh, CAMBRIDGE

Reference Books:

1) Foundations of Software testing, Aditya P Mathur, 2ed, Pearson
2) Software testing techniques – BarisBeizer, Dreamtech, second edition.
3) Software Testing, Principles, techniques and Tools, M G Limaye, TMH
4) Effective Methods for Software testing, Willian E Perry, 3ed, Wiley

E-Resources:

1) https://www.tutorialspoint.com/software_testing_dictionary/test_tools.htm